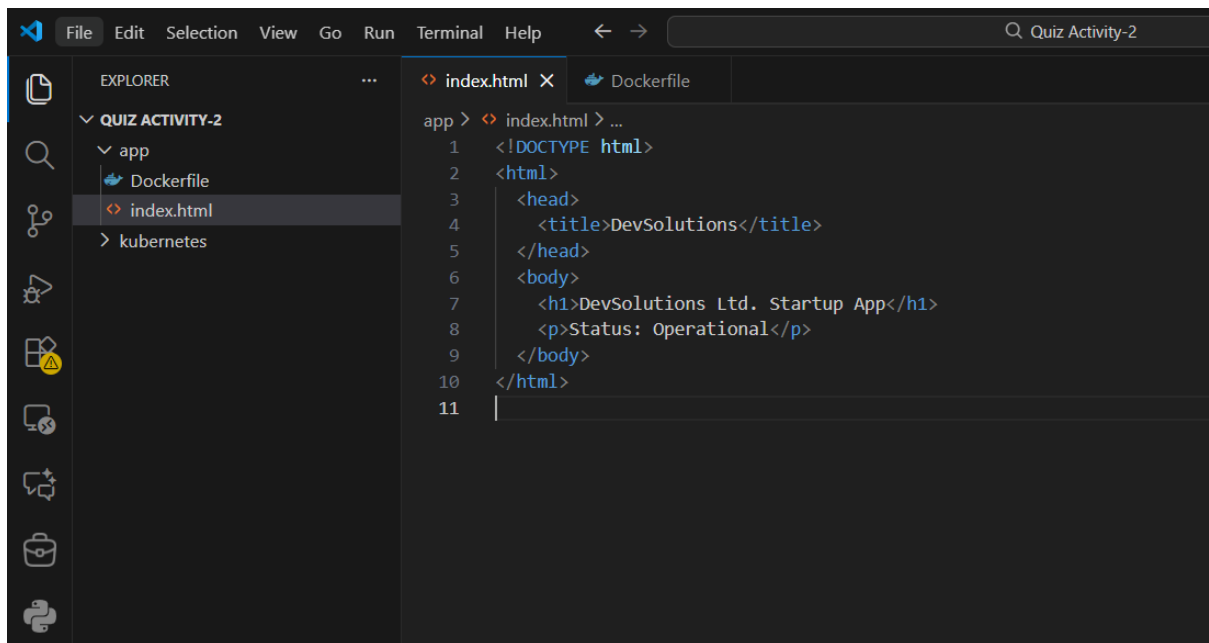
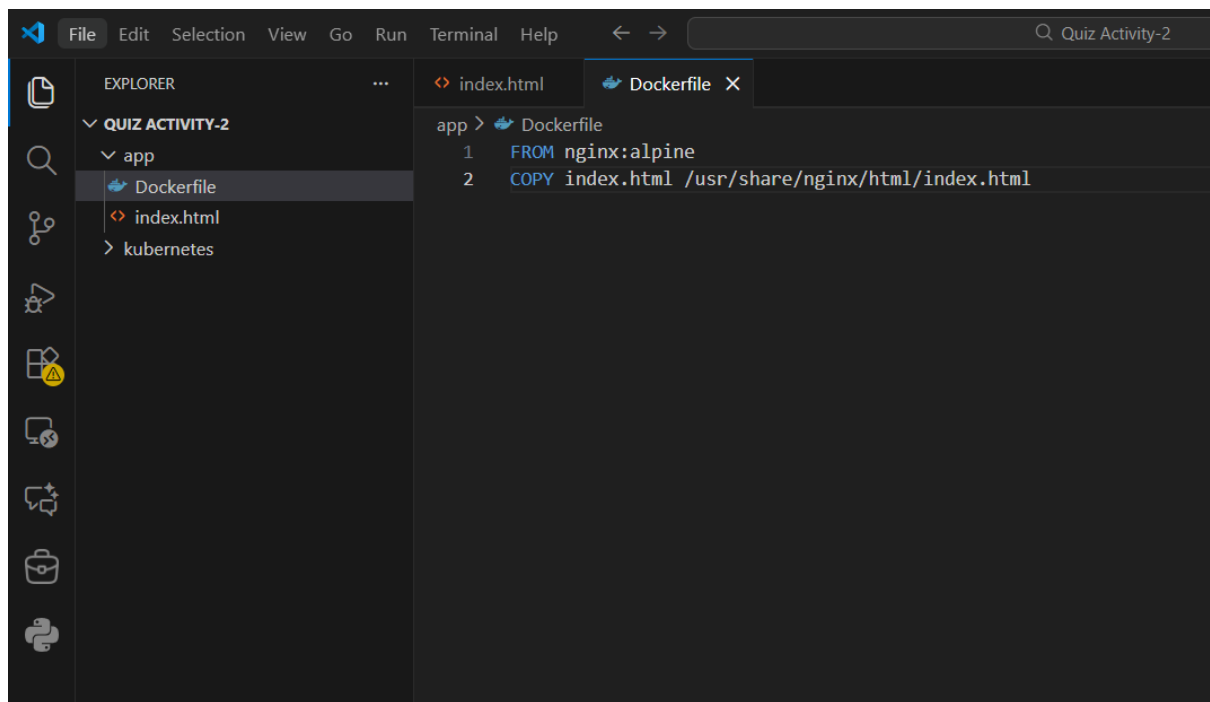


Phase 1: Container Bootcamp

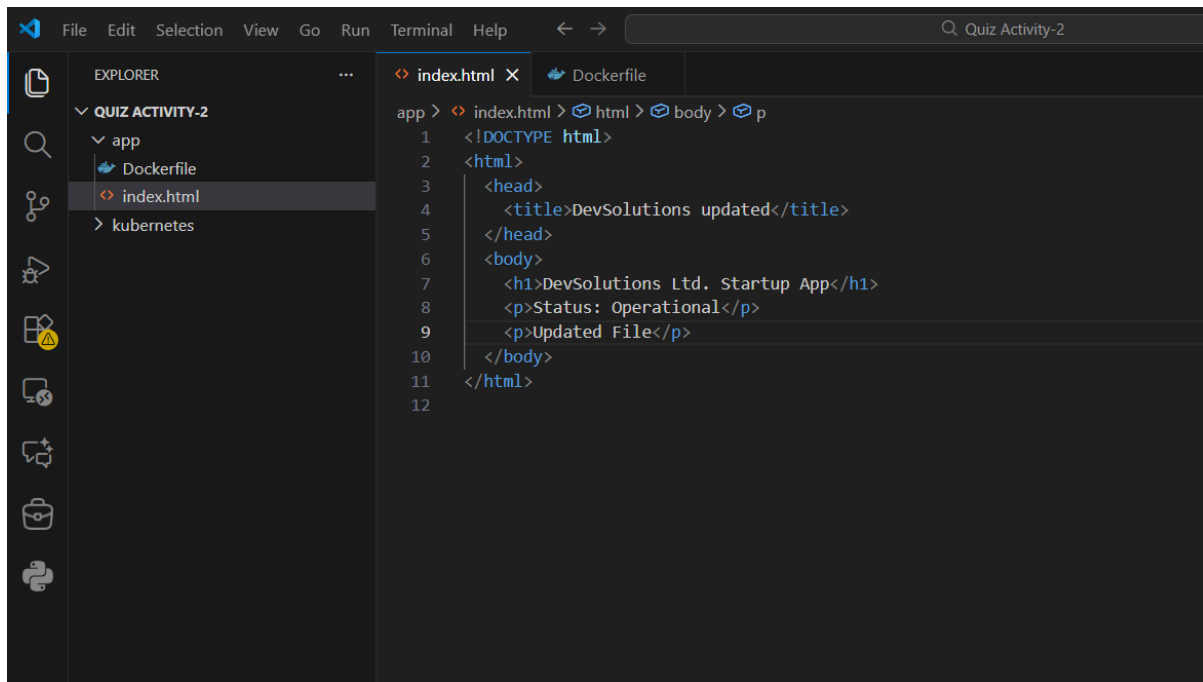


The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left displays a project structure under 'QUIZ ACTIVITY-2' with a subdirectory 'app' containing 'Dockerfile' and 'index.html'. The main editor area shows the 'index.html' file with the following HTML code:

```
app > index.html > ...
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>DevSolutions</title>
5   </head>
6   <body>
7     <h1>DevSolutions Ltd. Startup App</h1>
8     <p>Status: Operational</p>
9   </body>
10 </html>
11
```



```
Windows PowerShell
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app> docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:553f64aecdc31b5bf944521731cd70e35da4faed96b2b7548a3d8e2598c52a42
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app> docker run -d -p 80
80:80 --name manual-nginx nginx
fa9c45dee2b6a9c254f47028f07ca0ee2cf47a6d1099e4e4dfc62f7bf6fc8e01
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app> docker stop manual-
nginx
manual-nginx
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app> docker rm manual-ng
inx
manual-nginx
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app> docker build -t dev
solutions-web:v1 .
[+] Building 0.7s (7/7) FINISHED          docker:desktop-linux
=> [internal] load build definition from Dockerfile          0.0s
=> => transferring dockerfile: 104B                          0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine 0.0s
=> [internal] load .dockerignore                            0.0s
=> => transferring context: 2B                                0.0s
=> [internal] load build context                            0.1s
=> => transferring context: 224B                              0.0s
=> CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:b3c656d55 0.1s
=> => resolve docker.io/library/nginx:alpine@sha256:b3c656d55d7ad751 0.0s
=> [2/2] COPY index.html /usr/share/nginx/html/index.html 0.0s
=> exporting to image                                     0.3s
=> => exporting layers                                       0.1s
=> => exporting manifest sha256:436181abf9e6c5cf92baa998584e6b2ea594 0.0s
=> => exporting config sha256:4713a927dc02f3c4dbcdcbf151a38a7e98222ad 0.0s
=> => exporting attestation manifest sha256:93de4b9a72dc0d2f28e66f79 0.0s
=> => exporting manifest list sha256:39729c6b571c9bd0ea90e9cffe9b8e9 0.0s
=> => naming to docker.io/library/devsolutions-web:v1      0.0s
=> => unpacking to docker.io/library/devsolutions-web:v1    0.1s
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app> docker run -d -p 80
81:80 --name my-web-app devsolutions-web:v1
e66a660312f73a75a4f245c39ffda02213dd4fdc8dd773ee7269019c461ff041
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app>
```



```
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app> docker build -t dev
solutions-web:v2 .
[+] Building 0.5s (7/7) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile             0.0s
=> => transferring dockerfile: 104B                             0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine  0.0s
=> [internal] load .dockerignore                               0.0s
=> => transferring context: 2B                                    0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 257B                                  0.0s
=> CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:b3c656d55 0.0s
=> => resolve docker.io/library/nginx:alpine@sha256:b3c656d55d7ad751 0.0s
=> [2/2] COPY index.html /usr/share/nginx/html/index.html      0.0s
=> exporting to image                                           0.2s
=> => exporting layers                                           0.1s
=> => exporting manifest sha256:4c4de4fc78de2bc1d171cddd92871092e275 0.0s
=> => exporting config sha256:a45cbb44c0e79f4edb61afe8e8f272b5199ead 0.0s
=> => exporting attestation manifest sha256:1f76f27464c64ea7a564bf6a 0.0s
=> => exporting manifest list sha256:57f8166d71bf58bb78eb13111477a0e 0.0s
=> => naming to docker.io/library/devsolutions-web:v2           0.0s
=> => unpacking to docker.io/library/devsolutions-web:v2         0.0s
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app> docker run -d -p 80
82:80 devsolutions-web:v2
6b7321156f2b30536e0b74d2c916df8d385a0fdae1721d544e36c74dbfde82cf
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2\app>
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
6973211582b0	devolutions-web:v2	"/docker-entrypoint..."	29 seconds ago	Up 23 seconds	0.0.0.0:8082->80/tcp, [::]:8082->80/tcp	happy_hertz
64646031277	devolutions-web:v1	"/docker-entrypoint..."	3 minutes ago	Up 3 minutes	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp	my-web-app
632afce674e9	b3c656d55d7a	"/docker-entrypoint..."	3 minutes ago	Up 3 minutes		Myapp.devsolutions-app-69b6fd9967-bdhr.default.6a151ec1-92c8-4a91-a89e-3c81f97a76bc.2
4b9d2104058	b3c656d55d7a	"/docker-entrypoint..."	3 minutes ago	Up 3 minutes		Myapp.devsolutions-app-69b6fd9967-jc2n2.default.84df1613-2c1c-4619-85d8-c8a92a349fcd.2

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
devolutions-web	v2	57f31d66771bf39729c6b571c	About a minute ago	79.8MB
devolutions-web	v1	31b0b099580fe	2 days ago	79.8MB
myapp	latest	01d538cab09d	12 days ago	160MB
nginx	latest	553f64aecd0c3	2 weeks ago	225MB
postgres	13	46b9948c833	3 weeks ago	610MB
nginx	alpine	b3c656d55d7a	5 weeks ago	79.8MB
docker/desktop-kubemates	kubemates-v1.34.1-cni-v1.7.1-critools-v1.32.0-czi-dockerd-v0.3.20-1-debian	12d6d725d4e0	2 months ago	591MB
registry.k8s.io/kube-apiserver	v1.34.1	b9d7c117f8ac	2 months ago	110MB
registry.k8s.io/kube-scheduler	v1.34.1	6e9fbc0e25a5	2 months ago	73.9MB
registry.k8s.io/kube-controller-manager	v1.34.1	2b147c13b1f5	2 months ago	103MB
registry.k8s.io/kube-proxy	v1.34.1	913c83ca8b5	2 months ago	167MB
registry.k8s.io/coredns	1.6.4-deb	e3ac0816d3d2	4 months ago	272MB
registry.k8s.io/coredns/coredns	v1.12.1	e8c26256636	8 months ago	101MB
registry.k8s.io/pause	3.6	e80321729862	18 months ago	1.60MB
docker/desktop-apisil-controller	dc31cb22850b0cdd97c84a9cfecaf44a1af6e	76cf5576a070	2 years ago	470B
docker/desktop-storage-provisioner	v2.0	115d77efe6e2	4 years ago	59.2MB

▼

DevSolutions

×

+

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→

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ⓘ

localhost:8081

DevSolutions Ltd. Startup App

Status: Operational

▼

DevSolutions

×

DevSolutions updated

×

+

←

→

↺

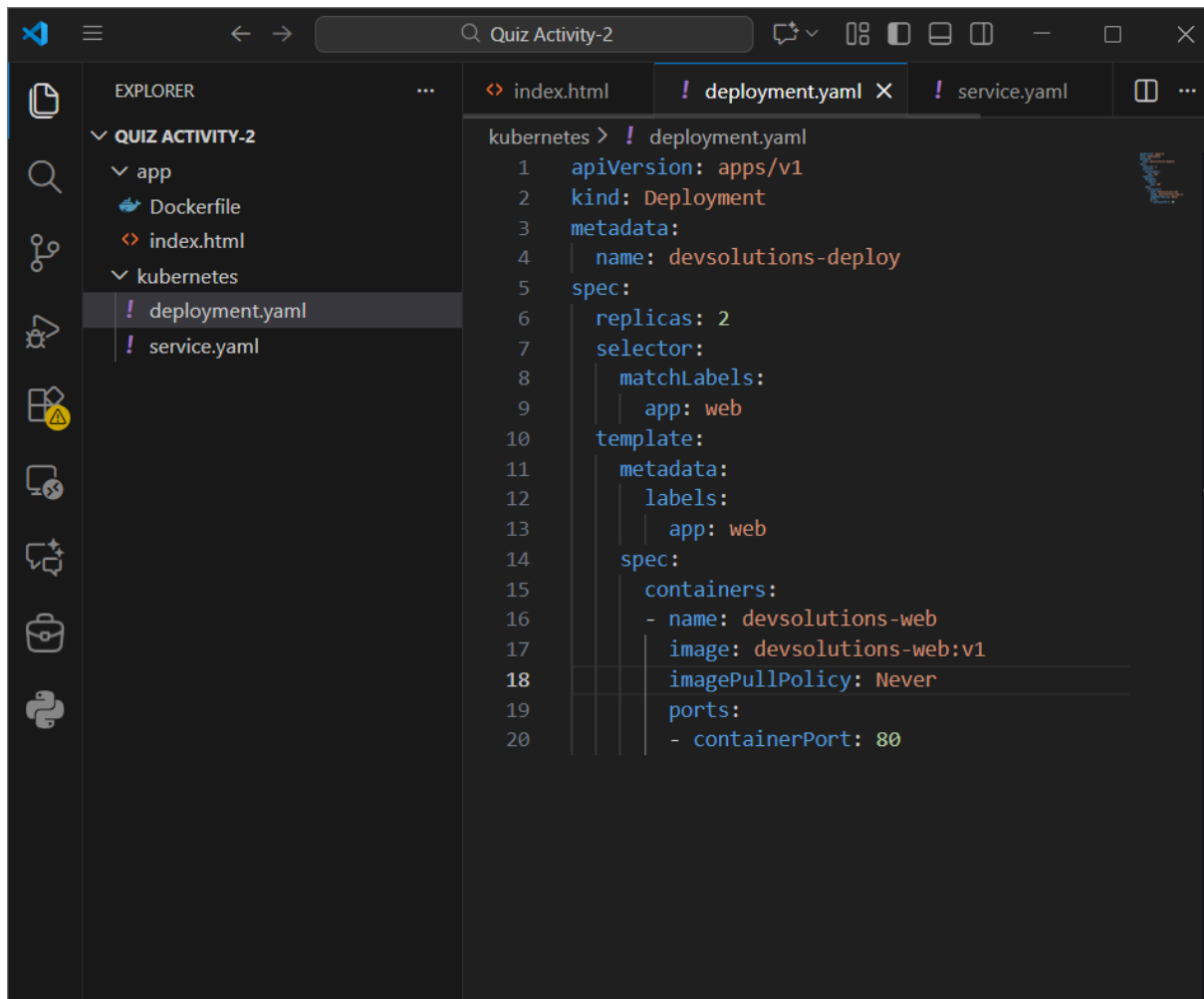
ⓘ

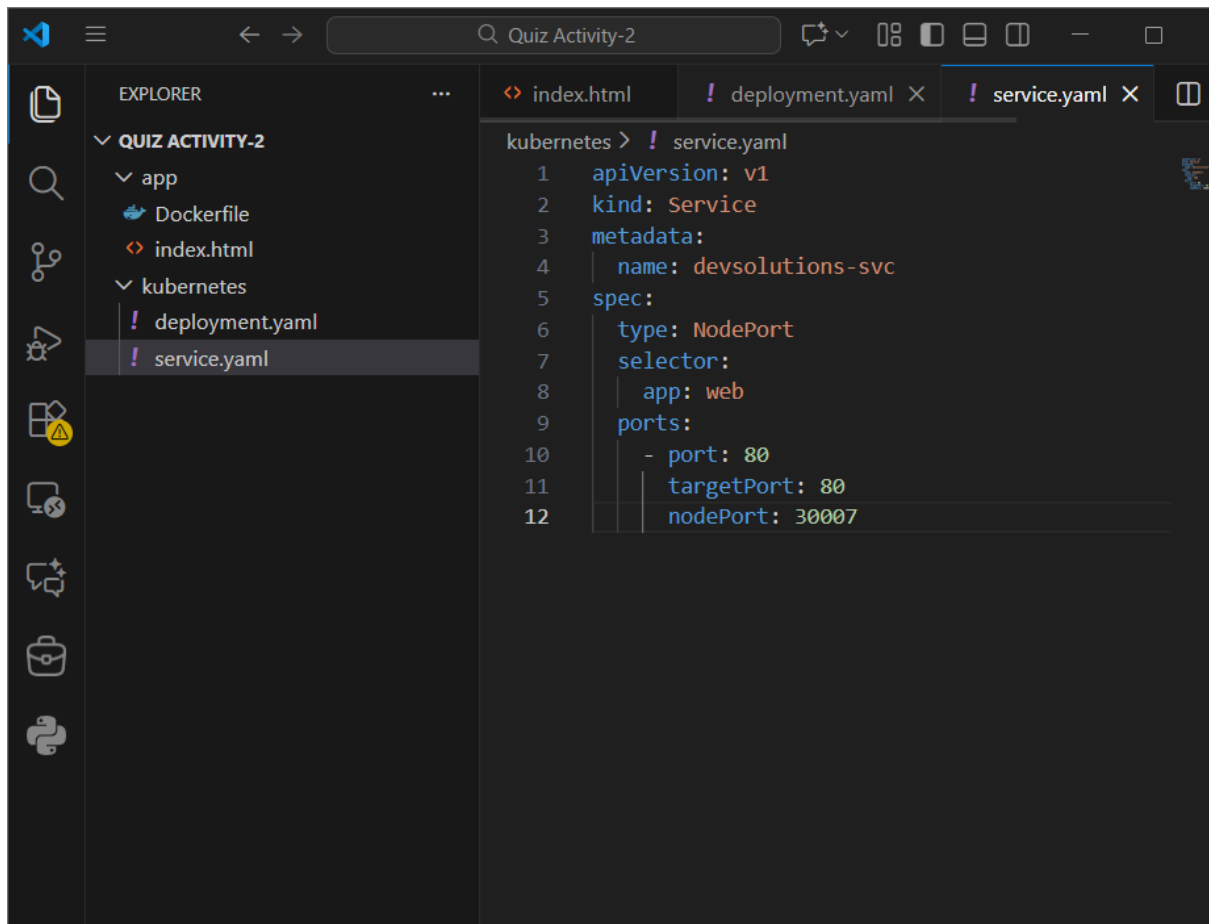
localhost:8082

DevSolutions Ltd. Startup App

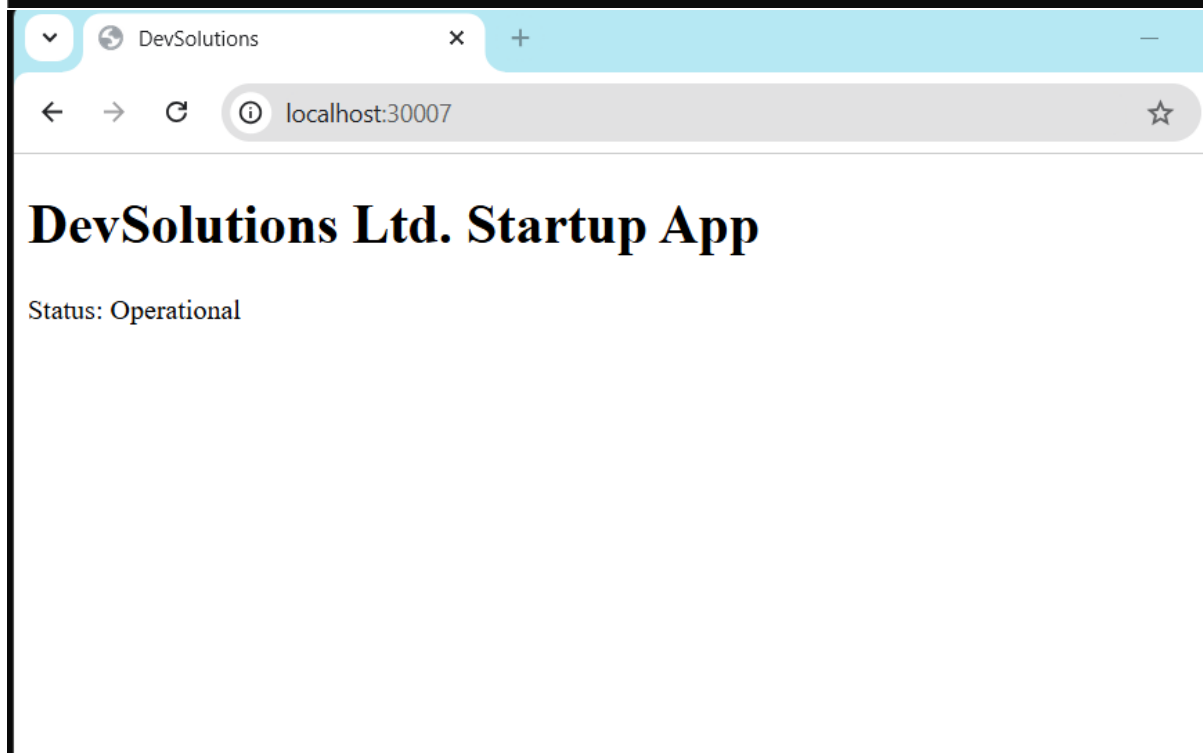
Status: Operational

Updated File

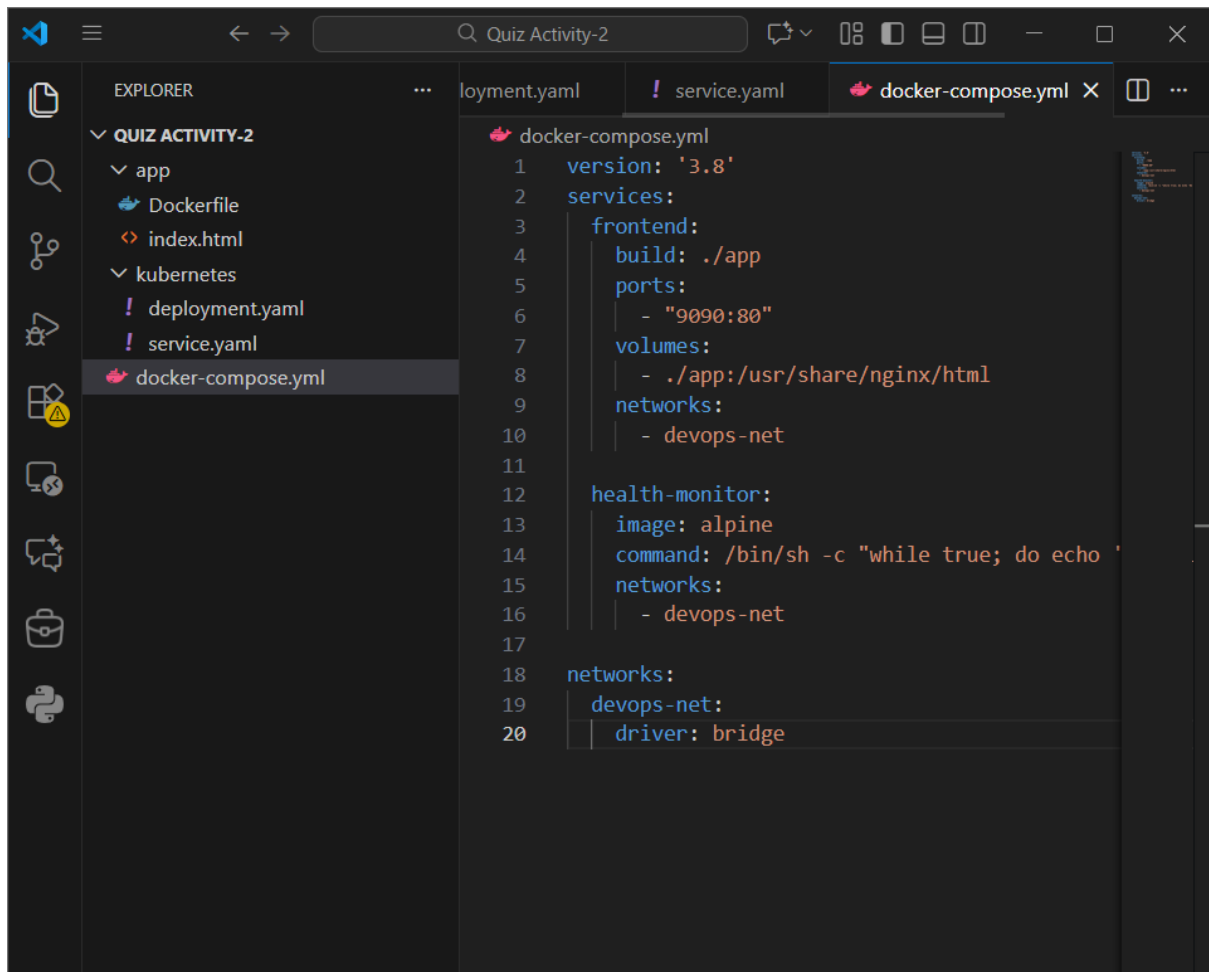




```
Windows PowerShell
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> kubectl apply -f kubernetes/deployment.yaml
deployment.apps/devsolutions-deploy unchanged
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> kubectl apply -f kubernetes/deployment.yaml
deployment.apps/devsolutions-deploy unchanged
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
devsolutions-app-69bcfd9967-bdhbr   1/1     Running   2 (18m ago) 3d9h
devsolutions-app-69bcfd9967-jk2n2   1/1     Running   2 (18m ago) 3d9h
devsolutions-deploy-86977f8497-62qvj 1/1     Running   0           7m48s
devsolutions-deploy-86977f8497-b2vzl 1/1     Running   0           7m48s
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> kubectl get svc
NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)        AGE
devsolutions-svc                   NodePort    10.103.5.245 <none>        80:30007/TCP   83s
kubernetes                         ClusterIP   10.96.0.1    <none>        443/TCP        25d
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> |
```



Phase 3: Multi-Service App



```

PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> docker-compose up -d --
build
time="2025-12-04T00:16:32+05:00" level=warning msg="D:\\Semester 5\\SCD\\Cla
ss Activity\\Quiz Activity-2\\docker-compose.yml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion
"
[+] Running 1/1
  ✓health-monitor Pulled                                4.6s
[+] Building 0.6s (9/9) FINISHED
=> [internal] load local bake definitions                0.0s
=> => reading from stdin 590B                          0.0s
=> [internal] load build definition from Dockerfile      0.0s
=> => transferring dockerfile: 104B                    0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine 0.0s
=> [internal] load .dockerignore                        0.0s
=> => transferring context: 2B                         0.0s
=> [internal] load build context                        0.0s
=> => transferring context: 32B                       0.0s
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:b3c656d55d7ad751 0.0s
=> => resolve docker.io/library/nginx:alpine@sha256:b3c656d55d7ad751 0.0s
=> CACHED [2/2] COPY index.html /usr/share/nginx/html/index.html 0.0s
=> exporting to image                                  0.1s
=> => exporting layers                                0.0s
=> => exporting manifest sha256:74ef93861392d148b02f061c4416e37fa8e6 0.0s
=> => exporting config sha256:8e1a12160d39582c519f41a9c0d09d654f8169 0.0s
=> => exporting attestation manifest sha256:ea76dd2b8e77716de432b969 0.0s
=> => exporting manifest list sha256:2f03f785203985d254e8920260d36b7 0.0s
=> => naming to docker.io/library/quizactivity-2-frontend:latest 0.0s
=> => unpacking to docker.io/library/quizactivity-2-frontend:latest 0.0s
=> resolving provenance for metadata file              0.0s
[+] Running 4/4
  ✓quizactivity-2-frontend                               Built           0.0s
  ✓Network quizactivity-2_devops-net                    Created         0.0s
  ✓Container quizactivity-2-health-monitor-1            Started        0.4s
  ✓Container quizactivity-2-frontend-1                  Started        0.5s
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> docker ps

```

```

PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> docker kill quizactivit
y-2-frontend-1
quizactivity-2-frontend-1
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> docker-compose start fr
ontend
time="2025-12-04T00:18:50+05:00" level=warning msg="D:\\Semester 5\\SCD\\Cla
ss Activity\\Quiz Activity-2\\docker-compose.yml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion
"
[+] Running 1/1
  ✓Container quizactivity-2-frontend-1 Started          0.2s
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> docker-compose ps
time="2025-12-04T00:19:01+05:00" level=warning msg="D:\\Semester 5\\SCD\\Cla
ss Activity\\Quiz Activity-2\\docker-compose.yml: the attribute `version` is
obsolete, it will be ignored, please remove it to avoid potential confusion
"

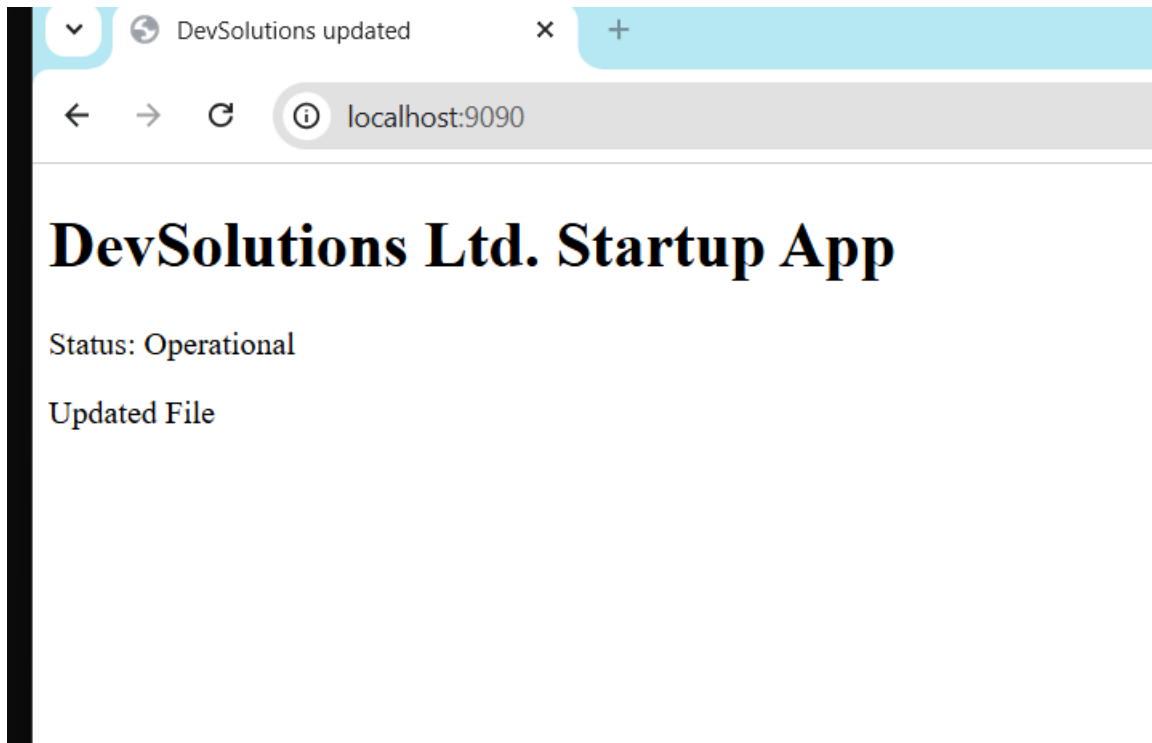
```

NAME	SERVICE	IMAGE	CREATED	STATUS	COMMAND
quizactivity-2-frontend-1	frontend	quizactivity-2-frontend	2 minutes ago	Up 11 seconds	"/docker-entrypo int...." frontend
quizactivity-2-health-monitor-1	health-monitor	alpine	2 minutes ago	Up 2 minutes	"/bin/sh -c 'whi le t..."

```

PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2>

```



Phase 4: CI/CD War Zone

Quiz Activity-2

EXPLORER

- QUIZ ACTIVITY-2
 - app
 - Dockerfile
 - index.html
 - kubernetes
 - deployment.yaml
 - service.yaml
 - docker-compose.yml
 - Jenkinsfile

Jenkinsfile

```

1 pipeline {
2   agent any
3   stages {
4     stage('Build Image') {
5       steps {
6         sh 'docker build -t devsolutio
7       }
8     }
9     stage('Test') {
10      steps {
11        echo 'Testing HTML syntax...'
12        sh 'ls app/index.html'
13      }
14    }
15    stage('Deploy to K8s') {
16      steps {
17        sh 'kubectl apply -f kubernet
18        sh 'kubectl rollout restart de
19      }
20    }
21  }
22 }
  
```

Windows PowerShell

```

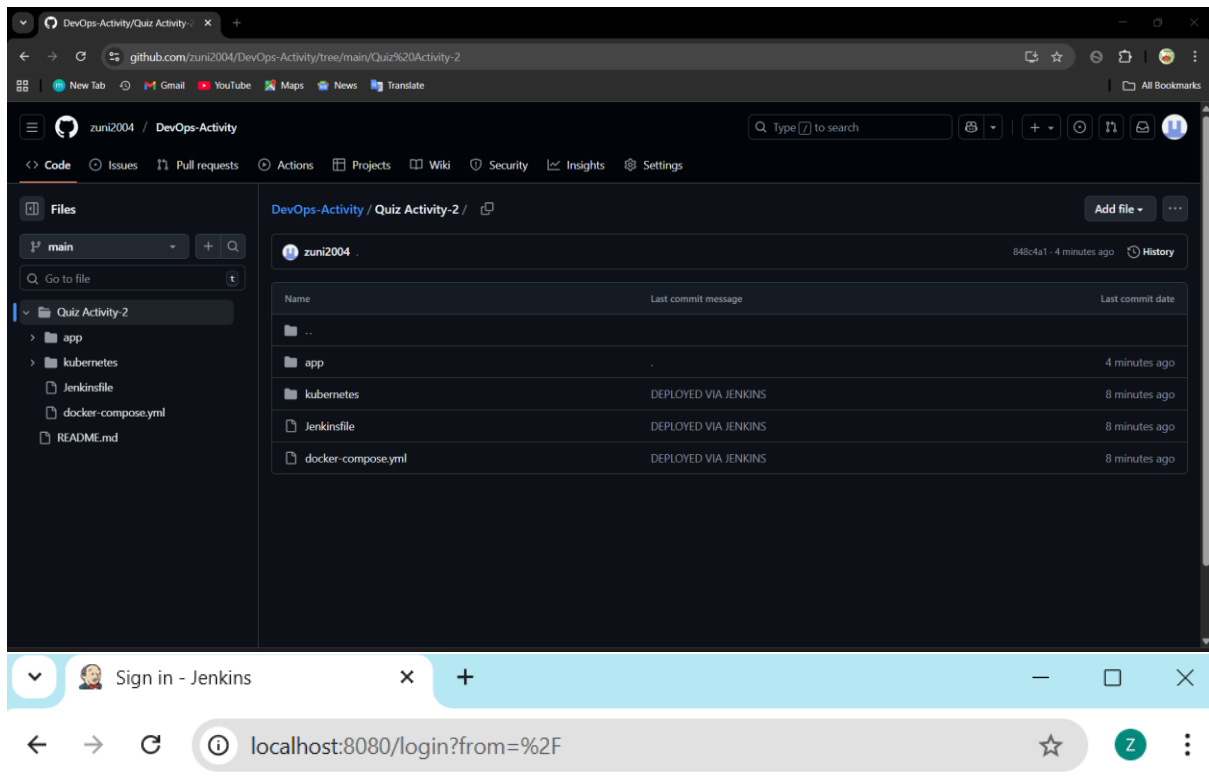
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> docker run -d -p 8080:8080 -p 50000:50000 --name jenkins -v /var/run/docker.sock:/var/run/docker.sock -
v jenkins_home:/var/jenkins_home jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
05a7d9a8b608: Pull complete
a37b858bb47a: Pull complete
744b4792e083: Pull complete
dc2a77f462ea: Pull complete
7718ff514022: Pull complete
c27509c3e53b: Pull complete
65e4ba8066bc: Pull complete
5dc07232677a: Pull complete
8d2a75b252b2: Pull complete
13cc39f8244a: Pull complete
e44beac64dffa: Pull complete
33300af18dd0: Pull complete
Digest: sha256:7b1c378278279c8688ef6d6168c25a1c2723a6bd6f0420beb5ccefabee3cc3bb1
Status: Downloaded newer image for jenkins/jenkins:lts
e46bbc970605accf393e51f4d81809e0e7eb55c5ed2de4552ade87cd75de8a5
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
e46bbc970605	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	12 seconds ago	Up 6 seconds	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
1e99708bac6f	alpine	"/bin/sh -c 'while t..."	9 minutes ago	Up 9 minutes	
40df5b5652dc	quizactivity-2-frontend	"/docker-entrypoint..."	9 minutes ago	Up 7 minutes	0.0.0.0:9090->80/tcp, [::]:9090->80/tcp
f97b0585f9a7	39729c6b571c	"/docker-entrypoint..."	20 minutes ago	Up 20 minutes	
d0713f765186	39729c6b571c	"/docker-entrypoint..."	20 minutes ago	Up 20 minutes	
6b7321156f2b	devsolutions-web:v2	"/docker-entrypoint..."	27 minutes ago	Up 27 minutes	0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
e66a660312f7	devsolutions-web:v1	"/docker-entrypoint..."	30 minutes ago	Up 30 minutes	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
632afc647e49	b3c656d55d7a	"/docker-entrypoint..."	31 minutes ago	Up 30 minutes	
4b540194b558	b3c656d55d7a	"/docker-entrypoint..."	31 minutes ago	Up 30 minutes	

```

PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2>
  
```



Sign in to Jenkins

Username

Password

☐ Keep me signed in

Sign in

```
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2> docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
b513d4b9bd0f400790c98e2abfda489e
PS D:\Semester 5\SCD\Class Activity\Quiz Activity-2>
```

DevSolutions-Pipeline Config - X +

localhost:8080/job/DevSolutions-Pipeline/configure

Jenkins

DevSolutions-Pipeline Configuration

Configure

- General
- Triggers
- Pipeline**
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/zuni2004/DevOps-Activity.git

Credentials ?

- none - + Add

Advanced

Save Apply

Dashboard - Jenkins

localhost:8080

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

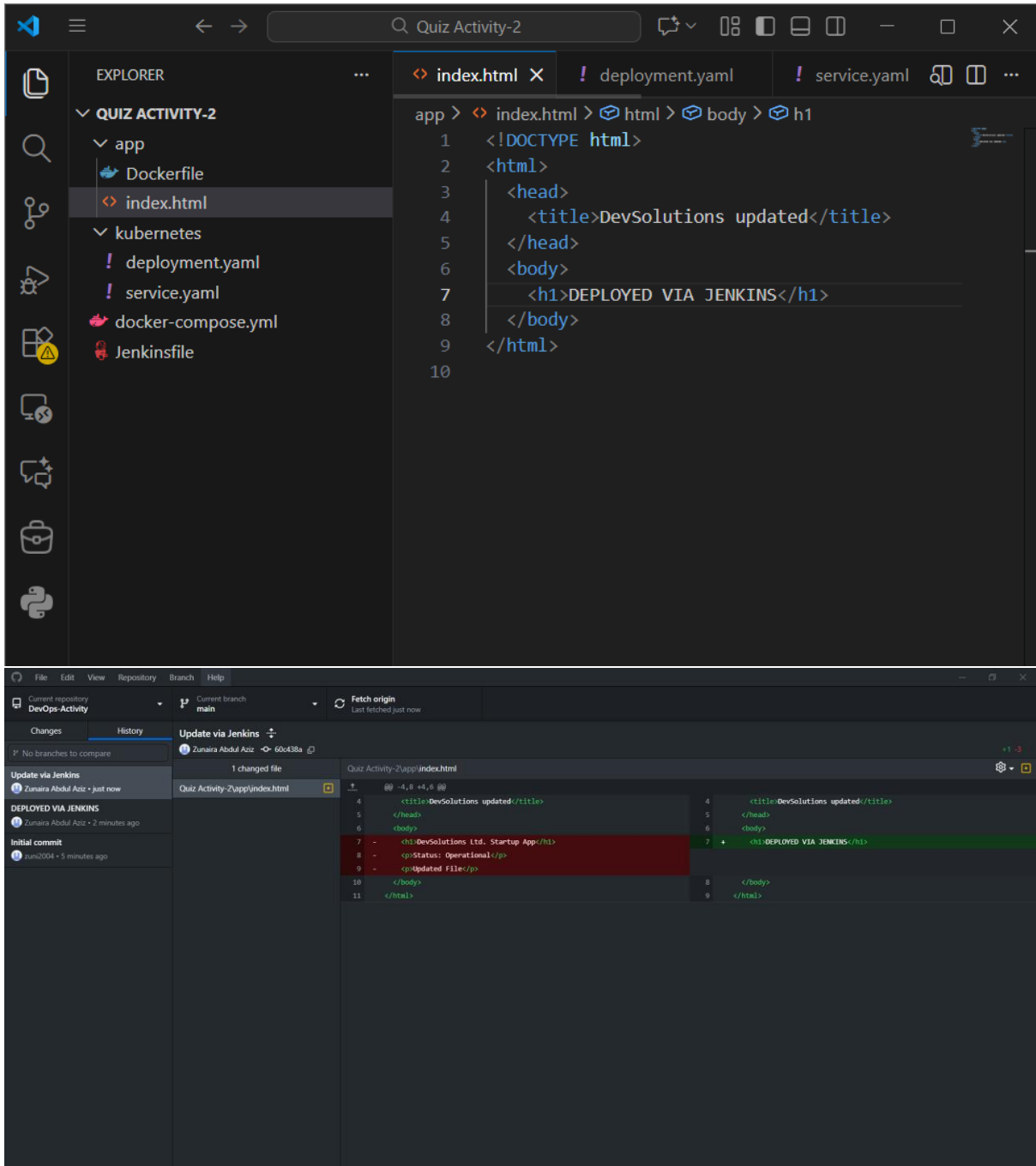
Build Executor Status

(0 of 2 executors busy)

All +

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	CI-CD-Lab5	25 days #10	N/A	9 min 28 sec
🔄	☀	DevSolutions-Pipeline	N/A	N/A	N/A

Icons: S M L



DevSolutions-Pipeline - Jenkins

localhost:8080/job/DevSolutions-Pipeline/

Jenkins

DevSolutions-Pipeline

Status

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

Credentials

DevSolutions-Pipeline

Add description

Stage View

Average stage times:
Full run time: ~9min 28s

Nov-08
14:16

No Changes

Setup: Install Docker	Create Project Files	Build	Test	Deploy to Kubernetes
9min 19s	204ms	4s	3s	421ms

Permalinks

- Last build (#5), 35 sec ago
- Last successful build (#5), 35 sec ago
- Last completed build (#6), 35 sec ago
- Last completed build (#5), 35 sec ago

Builds

Filter

Today

#5 7:38 PM

#4 7:38 PM

#3 7:37 PM